

Chapter 1

Sequences

LIMO 2026

Fix $p > 1$. Let $\{u_n\}_{n \in \mathbb{N}}$ be a sequence of positive numbers which satisfies for some constant $\alpha \in (0, 1)$ and any $n \in \mathbb{N} \setminus \{0\}$ the contraction property

$$|u_{n+1}^p - u_n^p| \leq \alpha |u_n^p - u_{n-1}^p|.$$

Prove that there exists $u_* \geq 0$ such that

(a)

$$\lim_{n \rightarrow \infty} u_n = u_*$$

(b) If $\alpha < \frac{1}{3}$, then

$$\sum_{n=0}^{\infty} |u_n - u_*| < \infty$$

IMC 2018, Day 2, Problem 7

Let $(a_n)_{n=0}^{\infty}$ be a sequence of real numbers such that

$$a_0 = 0, \quad a_{n+1}^3 = a_n^2 - 8 \quad (n \geq 0).$$

Prove that

$$\sum_{n=0}^{\infty} |a_{n+1} - a_n|$$

converges.

IMC 1998, Day 1, Problem 3

Let

$$f(x) = 2x(1 - x),$$

and let

$$f_n = \underbrace{f \circ f \circ \cdots \circ f}_{n \text{ times}}.$$

Find

$$\lim_{n \rightarrow \infty} \int_0^1 f_n(x) dx$$

and compute

$$\int_0^1 f_n(x) dx.$$

Putnam 2020, Problem A3

Let

$$a_0 = \frac{\pi}{2}, \quad a_n = \sin(a_{n-1}) \quad (n \geq 1).$$

Determine whether

$$\sum_{n=1}^{\infty} a_n^2$$

converges.

Putnam 2012, Problem B4

Suppose that

$$a_0 = 1, \quad a_{n+1} = a_n + e^{-a_n}, \quad n = 0, 1, 2, \dots$$

Does

$$a_n - \log n$$

have a finite limit as $n \rightarrow \infty$?

IMC 1996, Day 2, Problem 1

Let

$$f : [0, 1] \longrightarrow [0, 1]$$

be continuous, and let (x_n) be defined by

$$x_{n+1} = f(x_n).$$

Prove that (x_n) converges if and only if

$$\lim_{n \rightarrow \infty} (x_{n+1} - x_n) = 0.$$

IMC 2010, Day 2, Problem 1

(a) Let (x_n) be a sequence of real numbers satisfying

$$x_{n+1} = x_n \cos x_n.$$

Is (x_n) convergent for every choice of the initial value x_1 ?

(b) Let (y_n) be a sequence of real numbers satisfying

$$y_{n+1} = y_n \sin y_n.$$

Is (y_n) convergent for every choice of the initial value y_1 ?

Remark 1.1. *The contrast between parts (a) and (b) is instructive. In both cases,*

$$|x_{n+1}| \leq |x_n| \quad \text{or} \quad |y_{n+1}| \leq |y_n|,$$

so the absolute values always converge. However, in part (a), the choice $x_1 = \pi$ produces the persistent oscillation

$$\pi, -\pi, \pi, -\pi, \dots$$

In part (b), the oddness of the sine function together with the structure of the recurrence prevents such a nontrivial oscillation from persisting.